



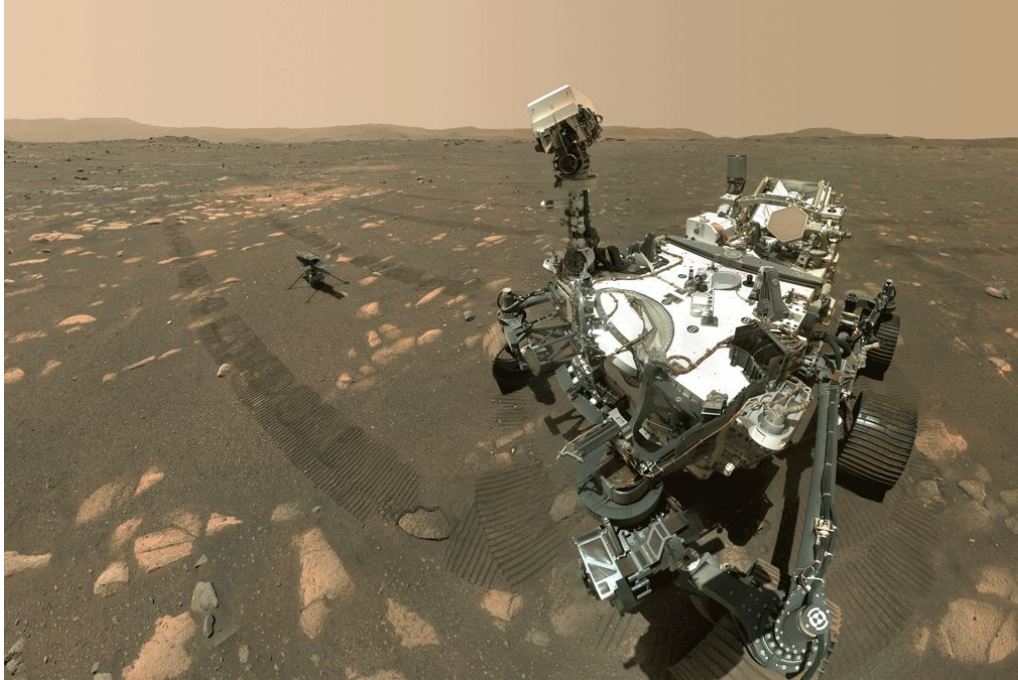
GRAN SASSO
SCIENCE INSTITUTE

Network-Aware Evaluation of Robotic Missions

A reusable evaluation workflow for distributed ROS 2 systems

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Network in (Remote) Robotic Missions



Perseverance Rover and Ingenuity — Image from NASA

Earth-to-Mars Latency

Communication with Earth control stations is mandatory during long traverses and can take up to **22-minutes one-way**.

Without it, local location estimation drift can exceed 35 meters.

Onboard Compute Optimization

Integrating repurposed onboard compute hardware inspired by the Ingenuity Mars Helicopter removes the day-long ground loop bottlenecks.

Reduces localized error estimation down to 25 cm with computation done within 2-minutes.

Space It Up! Project

The **Space It Up!** project is funded by the **Italian Space Agency (ASI)** and the **Ministry of University and Research (MUR)**.

GSSI is working on developing and evaluating meaningful testing scenarios and environments for Robotics Space Exploration.

Network in (Remote) Robotic Missions

Autonomous Underwater Vehicles (AUVs) can use real-time ocean-current data and forecasts to adapt their sampling strategies.

Energy Optimized Routes

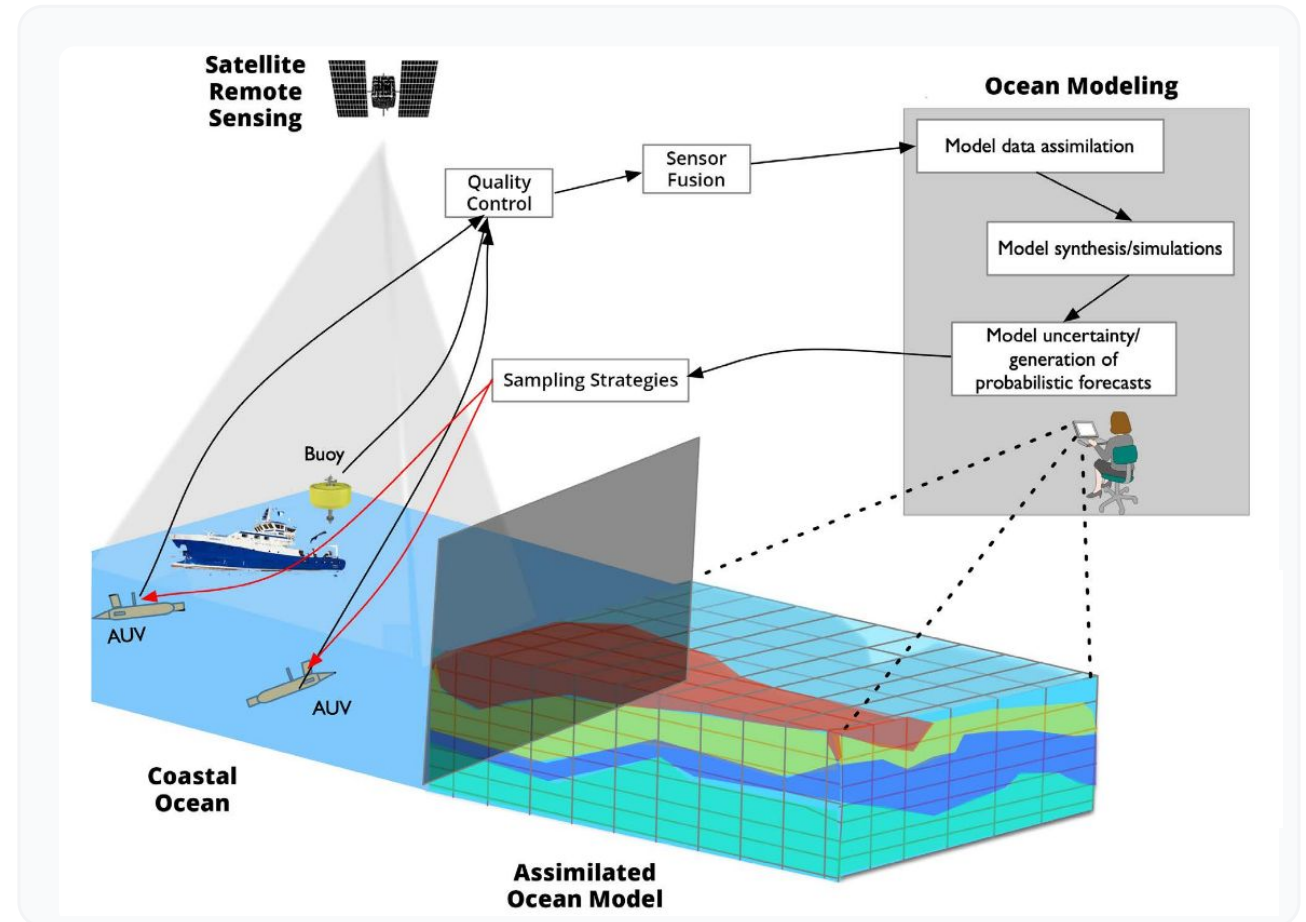
Utilize real-time ocean current updates to plan highly efficient navigation paths.

Cloud-Based Planning

Offload complex data assimilation and intensive computational planning tasks to the cloud.

Idle Time Management

Optimize idle periods to significantly reduce **energy** consumption and improve vehicle **safety**.

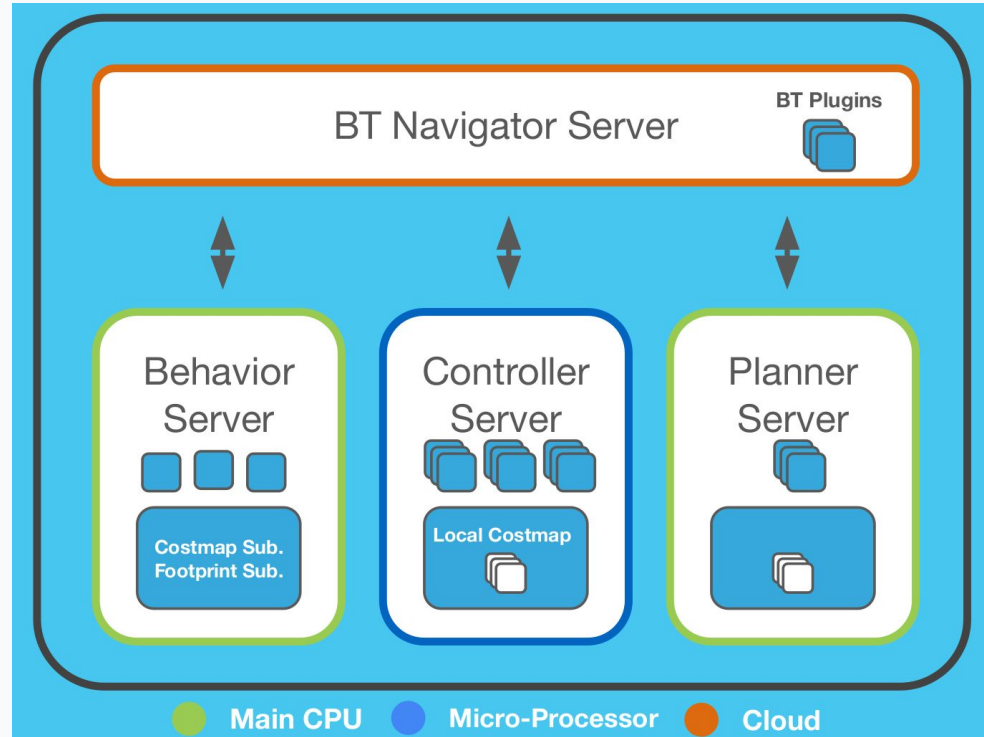


From: Mendes, Renato, et al. "TOWARD OCEAN MODEL-DRIVEN." *Oceanography* 39.2 (2026).

Networks in ROS 2 Systems

Time-Sensitive Control

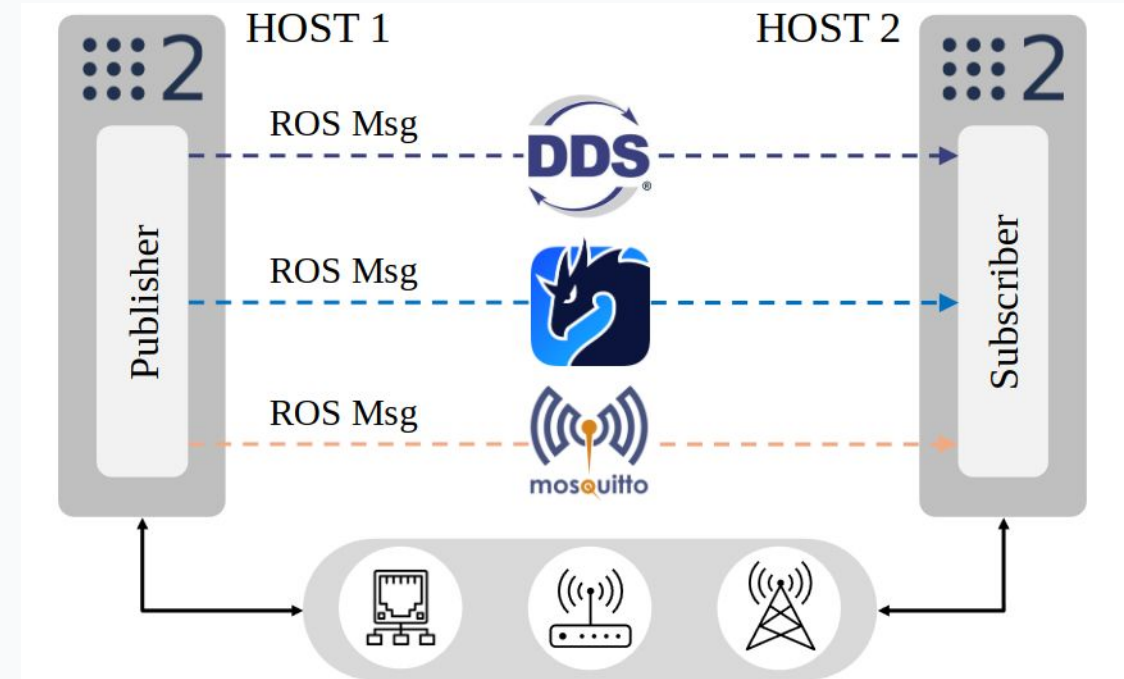
Low-latency feedback loops assumptions for smooth paths generation



From: Open Navigation

Heterogeneous Data Links

Integration with diverse physical communication channels.

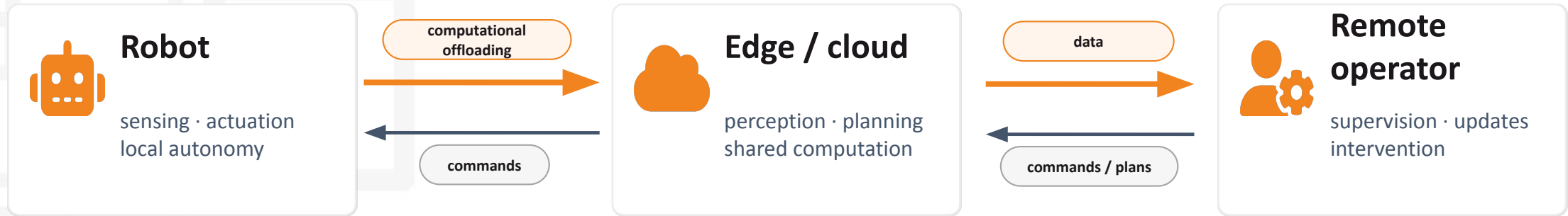


From: Zhang, Jiaqiang. "Robotic capabilities in distributed systems: Communication, localization, and resilience", University of Turku, 2026.

KEY Problem

Once perception or control is distributed, communication infrastructure and nodes placement is no longer negligible. It becomes part of the mission quality.

Network-Aware Evaluation of Robotic Missions



Higher-Level

Analysis of mission-critical nodes and key data flows driving robotic system behaviors.



Reuse Barrier

Most robotic co-simulations are highly application-specific and difficult to adapt or reuse.



Lower-Level

Evaluations focus on message routing and network topologies in MRS contexts.

Goal: operational requirements for the network infrastructure, node placement, and distributed deployment architecture

Research Questions

RQ1

Network Considerations

How does network infrastructure impacts missions with distributed robotic components?

RQ2

Nodes Distribution

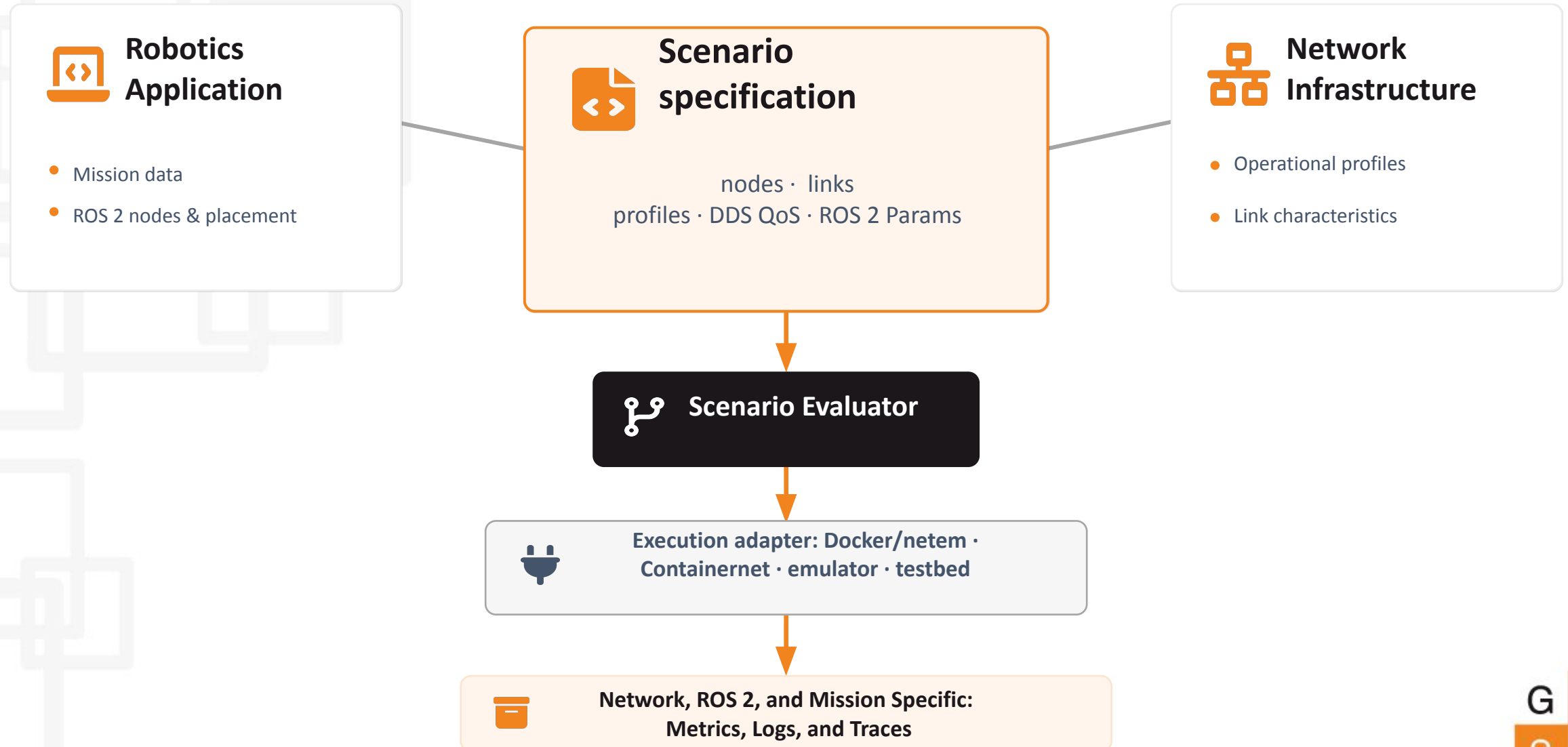
How can critical ROS 2 components be automatically extracted and evaluated considering network and placement options?

RQ3

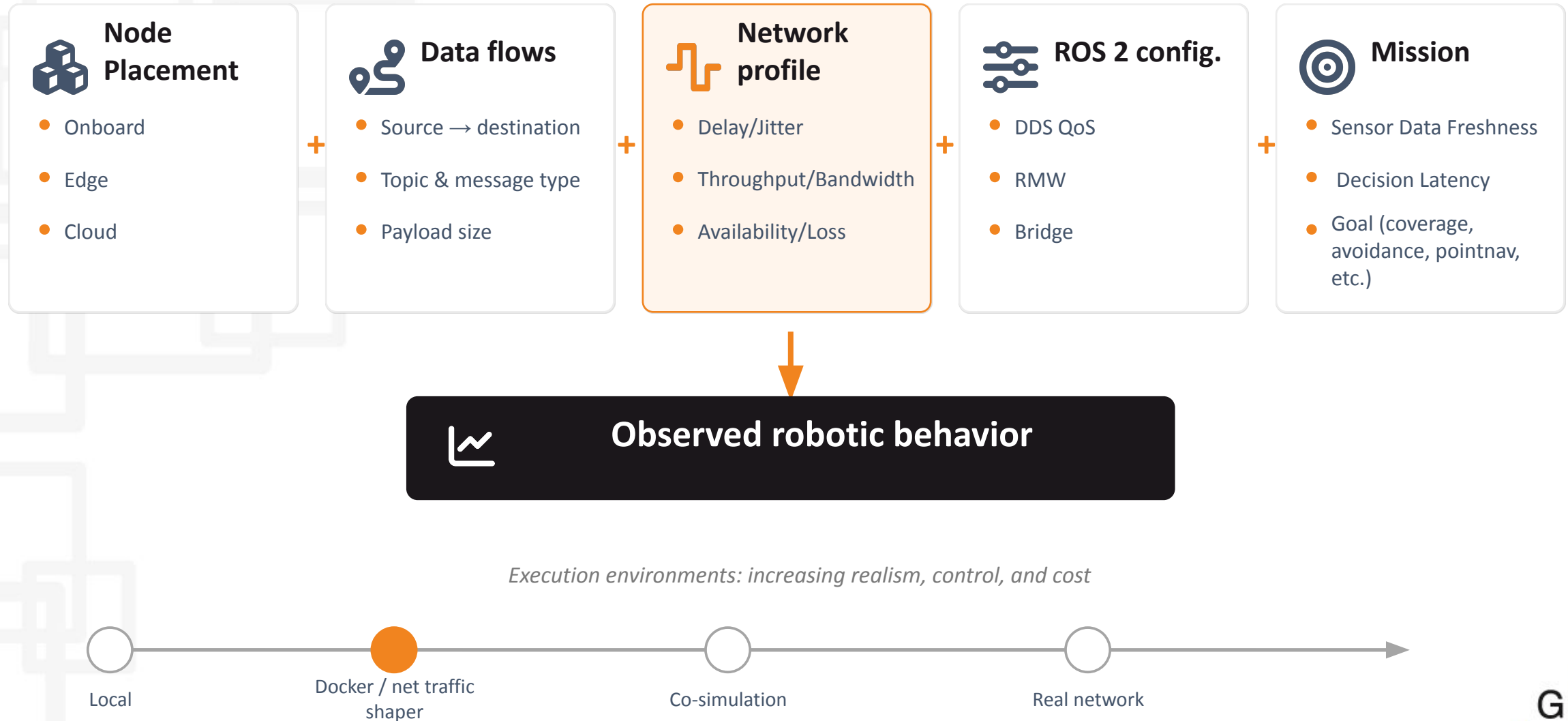
Reusability

How can network-aware ROS 2 systems evaluation be reused in different contexts?

Network-Aware Evaluation of Robotic Missions



Network-aware evaluation configuration space



Evaluation Parameters: placement × network profile × ROS 2 QoS

1 · Node placement

Where does each ROS 2 Node?

Onboard

Edge

Cloud

2 · Network operational profile

What are the network characteristics for the infrastructure options?

WiFi

Celular (4G)

Celular (5G)

Satellite

3 · ROS 2 QoS policy — configured per flow

QoS Policy

Parameters

Reliability

BEST_EFFORT

RELIABLE

History + depth

KEEP_LAST (N)

KEEP_ALL

Durability

VOLATILE

TRANSIENT_LOCAL

Timing / freshness

DEADLINE

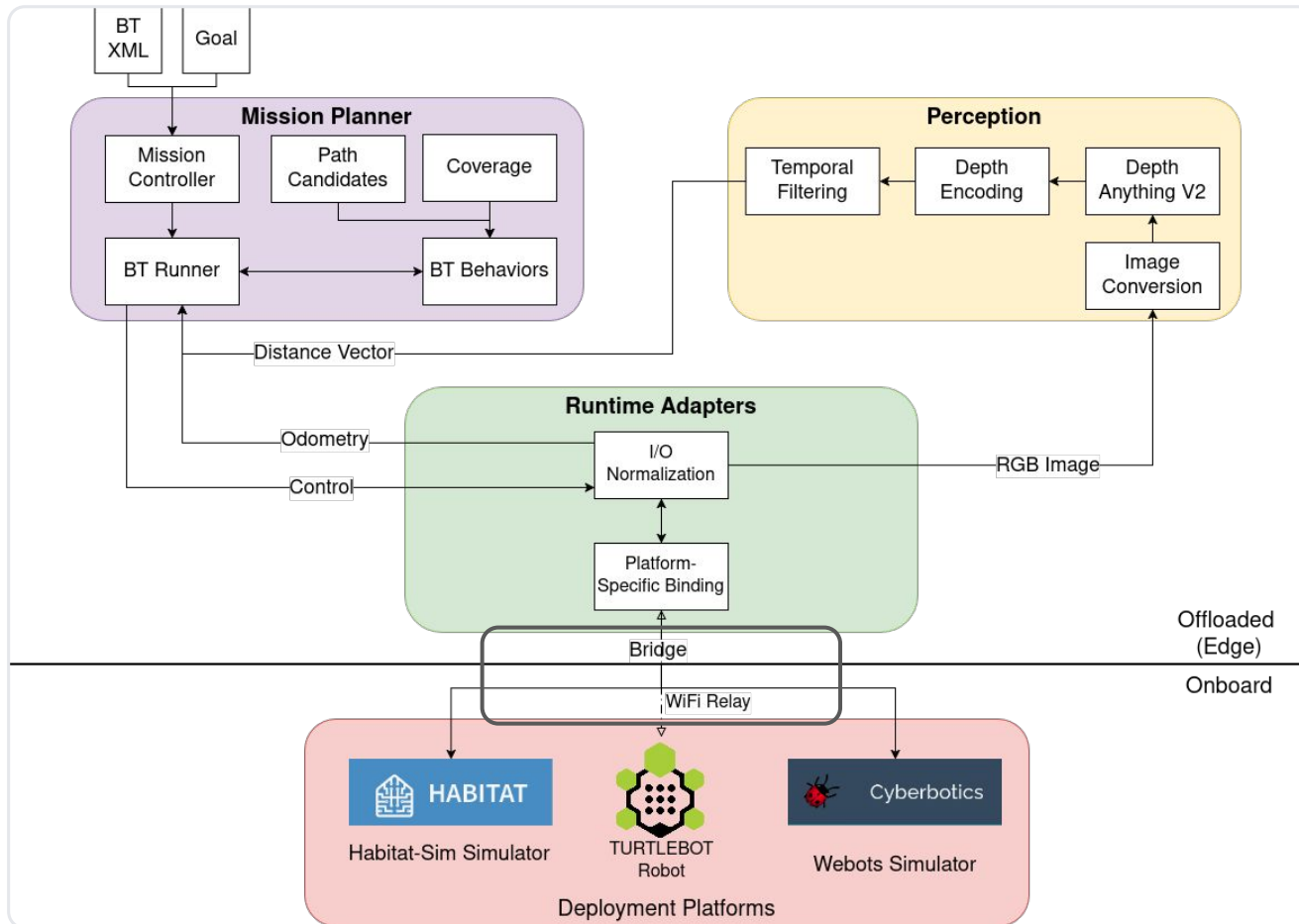
LIFESPAN

Liveliness

AUTOMATIC

MANUAL_BY_TOPIC + lease

Planned Proof-of-Concept: CogNav Edge-Offload Experiment



Navigation based on Monocular Depth Estimation



Uplink (Large Data Flow)

Offloaded depth computation from camera images



Downlink (Control Messages)

Commands onboard robot requiring small downlink

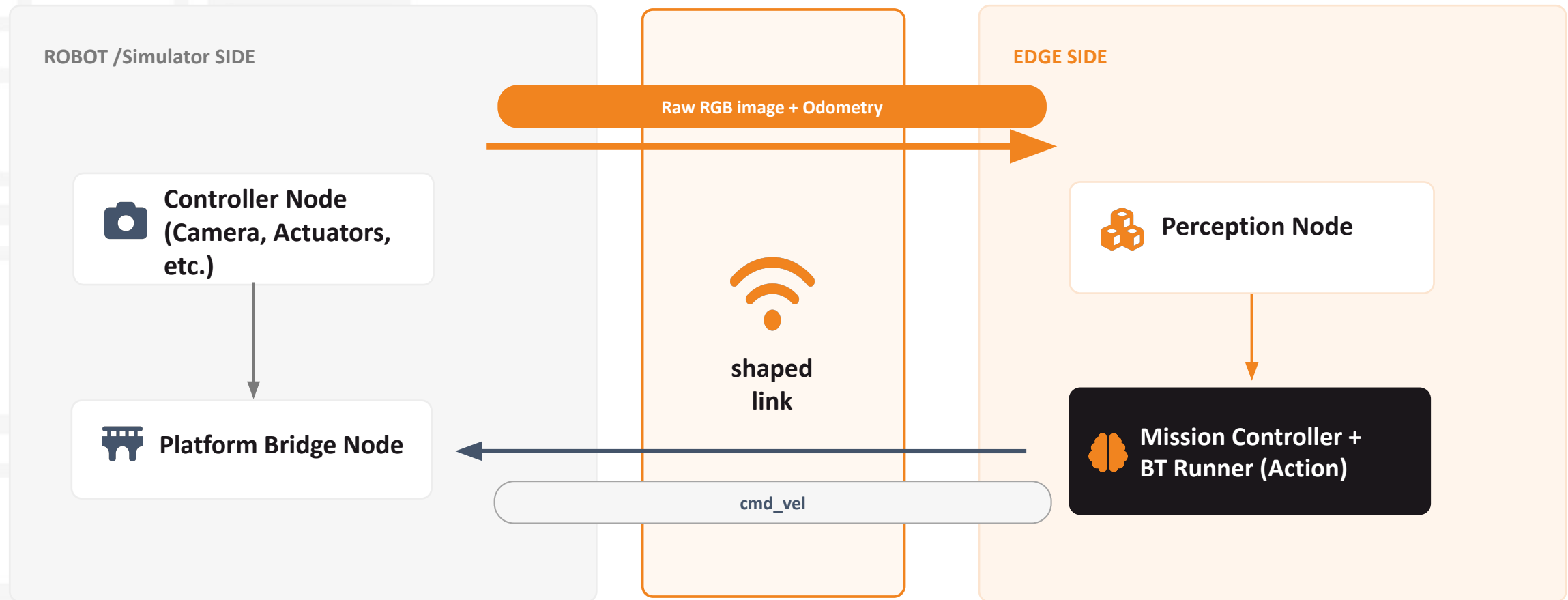


Real-World Validation

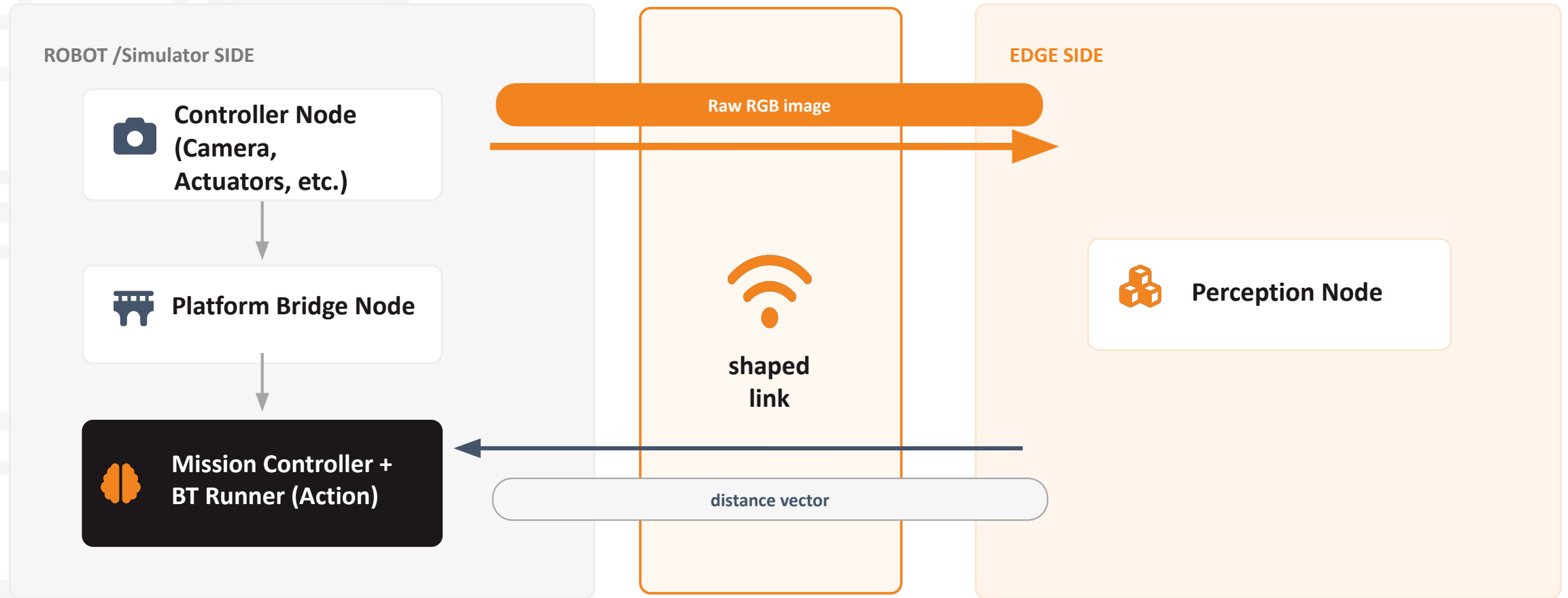
Compare simulated Wi-Fi profile with evaluation in real-world setup

Architecture under refactoring adapted from: Nikmard, M., Pelliccione, P., and D'Angelo, G.. "CogNav: Human-Inspired Collision Avoidance Strategy". In: 8th International Workshop on Robotics Software Engineering (RoSE'26). 2026

CogNav Edge-Offload Experiment: Placement Option 1 (current)



CogNav Edge-Offload Experiment: Placement Option 2



Future Work



Placement Alternatives

Understand how can semantic valid placement alternatives can be selected.



Multi-Segment Links

Analyze performance and routing across multi-segment link topologies.



Data Compression

Add data compression techniques into the active evaluation mix.



Energy Consumption

Incorporate key power and energy consumption considerations.



CI Automation

Implement automated continuous integration for the evaluation pipeline.



VPN Unified Network

Consider VPN architectures to place all nodes in the same network.

Thank You!